

RESEARCH

Open Access



Artificial intelligence and telemedicine in elderly healthcare: A mixed-methods study

Sameen Rafi^{1*} 

Abstract

Background Artificial Intelligence (AI) and telemedicine are increasingly integrated into geriatric healthcare, offering opportunities for early detection, monitoring, and improved access. Adoption among older adults in India remains limited because of accessibility, digital literacy, cost, and ethical concerns.

Methods A cross-sectional mixed-methods study was conducted in Aligarh (urban, semi-urban), India, between January and June 2025. Quantitative surveys were administered to 200 older adults (≥ 60 years) and in-depth interviews were conducted with 20 elderly participants and 10 healthcare professionals. Quantitative data were analyzed using descriptive statistics, correlation, and multiple regression; qualitative data underwent thematic analysis.

Results Fifty-two percent of participants reported using at least one AI-enabled healthcare tool (wearable monitors 32%; teleconsultations 28%). Adoption was higher in urban than rural participants ($p < 0.01$). Regression analysis showed digital literacy ($\beta = 0.41$, $p = 0.002$) and family support ($\beta = 0.36$, $p = 0.004$) were significant predictors of perceived empowerment, independent of age and education. Reported benefits included convenient access (61%), improved chronic-disease monitoring (54%), and better medication adherence (42%); primary barriers were low digital literacy (49%), cost (45%), and lack of trust (37%). Healthcare professionals highlighted data-privacy and ethical concerns and the risk of reduced human contact.

Conclusions AI-enabled telemedicine holds promise for improving aspects of geriatric care in India but is constrained by inequities, literacy gaps, affordability, and ethical challenges. Policy actions to promote elder-centered digital literacy, subsidized access, and regulatory safeguards are needed to ensure technologies complement rather than replace human care. Findings may inform national digital-inclusion and elder-care initiatives.

Keywords Artificial intelligence, Telemedicine, Elderly healthcare, Digital literacy, Family support, Geriatric empowerment

*Correspondence:

Sameen Rafi
sameenrafi156@gmail.com

¹Aligarh Muslim University, Aligarh, India



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Background

The global demographic shift toward aging populations presents urgent challenges for healthcare systems. By 2050, the number of people aged 60 years and older will exceed 2.1 billion, with the fastest growth occurring in low and middle-income countries [29]. Older adults experience a high burden of multimorbidity, frailty, and cognitive decline, often requiring long-term and complex care [22]. Conventional healthcare models are insufficient to meet these needs, especially in settings with limited specialist availability and healthcare infrastructure. In this context, research has emphasized the importance of age-friendly communities that provide supportive environments for older adults to live independently and maintain well-being [19].

Artificial Intelligence (AI) and telemedicine are increasingly recognized as transformative tools in geriatric care. AI encompasses machine learning, predictive analytics, robotics, and natural language processing, which can support early disease detection, personalized care, and efficient resource allocation [4]. Telemedicine expands these capabilities by enabling remote consultations, continuous monitoring, and timely interventions, particularly for older adults with mobility restrictions [8]. Together, AI and telemedicine hold the potential to improve healthcare access, quality, and equity for aging populations.

Despite these advances, critical questions remain regarding adoption, effectiveness, and sustainability in elderly care. Studies often focus on specific devices or pilot interventions, with limited evidence from low and middle-income countries such as India. Barriers including digital literacy, affordability, algorithmic bias, and ethical concerns further constrain large-scale implementation [16].

While global studies have explored the role of AI and telemedicine in elderly care, evidence from low- and middle-income countries remains scarce. This study is among the first mixed-methods empirical investigations in India to examine adoption patterns of AI-enabled healthcare tools among older adults. Unlike prior research that often isolates technological or clinical aspects, our study uniquely integrates digital literacy, family support, and cultural caregiving norms as predictors of empowerment. By capturing both quantitative associations and qualitative lived experiences, this research provides a context-specific contribution to gerontechnology literature and highlights the sociocultural dimensions of technology adoption in aging societies.

Review literature

The accelerating growth of older populations has intensified demand for innovative, age-sensitive healthcare. Traditional systems struggle with multimorbidity, frailty,

and social isolation among older adults, prompting interest in AI and telemedicine as tools for early detection, continuous monitoring, and improved care coordination. Evidence spans wearable sensors, IoMT frameworks, socially assistive robots, and virtual interventions, but studies vary widely in design and scope.

Multimorbidity and chronic disease monitoring

Older adults frequently present multiple chronic conditions that benefit from remote monitoring. Wearable sensors and RPM platforms can continuously track physiological signals and support personalized management [21, 27]. Recent work layering AI onto RPM data shows promise for predicting short-term decompensation in cardiovascular disease and improving care triage, though integration, cost, and data-quality challenges remain [24, 25].

For geriatric syndromes intertwined with multimorbidity, frailty and falls are active targets. A 2025 review synthesizing studies up to 2024 concludes wearable-sensor frailty screening is feasible and useful for rapid, objective assessments; emerging work even shows wrist-worn gait biomarkers identifying frailty in large samples [30]. For geriatric syndromes intertwined with multimorbidity, frailty and falls remain key clinical targets. Recent systematic reviews demonstrate that wearable-sensor technologies offer feasible, objective approaches for frailty screening and fall-risk assessment in community and clinical settings. *Bonanno et al.* [2] mapped mHealth pipelines for fall prediction from sensing and data processing to risk stratification and highlighted the need for multimodal datasets and real-world validation. Complementary qualitative syntheses, such as *Li et al.* [12], emphasize user acceptability challenges, including device comfort, skin irritation, and loss or misuse, which can limit adherence and long-term engagement among older adults. Collectively, this evidence underscores that while sensor-based monitoring is technically mature, successful implementation depends on usability design and user-centered deployment strategies.

Cognitive decline and mental health support

Cognitive impairment, including dementia and Alzheimer's disease, is a central concern in gerontology. AI-driven tools such as neural networks and virtual assistants have been used to monitor activity patterns, provide cognitive exercises, and detect early signs of cognitive decline [15]. Similarly, AI-based chatbots and emotional recognition systems aim to alleviate loneliness, depression, and social withdrawal among older adults [9]. At the same time, elderly women in India continue to face significant physical and mental health challenges, underscoring the need for healthcare innovations that address both medical and psychosocial vulnerabilities [20].

Speech and behavior-based digital biomarkers and conversational AI offer scalable approaches to screening and psychosocial support. Studies report that acoustic linguistic models and passive sensing can help detect cognitive impairment earlier, while chatbots and voice assistants can reduce loneliness when well integrated with human care [1, 14, 15]. Yet sample representativeness and usability for older adults require further attention.

Rehabilitation and mobility assistance

Robotic exoskeletons, sensor-based platforms, and VR physiotherapy show gains in gait, balance, and functional outcomes in controlled settings [11, 13]. Their high costs and infrastructure demands, however, limit scalability for community-dwelling elders [18, 23].

Social isolation and companionship technologies

AI companions and robotic pets can improve psychosocial well-being, particularly in institutional settings [7, 25]. Successful deployments prioritize empathy, personalization, and privacy to enhance acceptability among older adults [10].

Ethical, legal, and implementation barriers

Across domains, digital literacy gaps, algorithmic bias, privacy risks, and limited legal protections constrain equitable deployment [6, 28]. Care providers report concerns about training and the potential erosion of therapeutic relationships if human-centred safeguards are not maintained [3, 5].

In sum, AI and telemedicine hold substantial potential for elderly care particularly for monitoring and access but the evidence is often limited to pilots or single-site studies. There is a clear need for geriatric-specific datasets, inclusive design, cost-sensitive models, and implementation research to ensure technologies advance equity and dignity in aging.

Objectives

1. To examine the patterns of adoption and usage of AI-enabled telemedicine tools among older adults in India.
2. To assess the impact of digital literacy and family support on elderly individuals' empowerment in using AI-based healthcare technologies.
3. To evaluate the perceived benefits and challenges of AI-driven healthcare solutions from both elderly participants and healthcare professionals.
4. To identify gaps and barriers technological, ethical, and social that influence the sustainable integration of AI into geriatric care.

Methodology

Study design

A cross-sectional mixed-methods design was adopted to examine the use of Artificial Intelligence (AI) and telemedicine in elderly healthcare. This approach captured both the breadth of adoption patterns through quantitative surveys and the depth of lived experiences through qualitative interviews.

Study setting and participants

The study was conducted in Aligarh city, Uttar Pradesh, India, between January and June 2025. Aligarh represents an urban–semi-urban mix, offering a diverse sociocultural and infrastructural context for understanding technology adoption among older adults.

Older adults aged 60 years and above were recruited through community health centers, local NGOs, and senior-citizen associations. A purposive sampling strategy ensured representation from both urban and semi-urban areas.

- *Quantitative sample:* 200 elderly participants who consented to complete the survey (mean age = 68.2 years, SD = 6.4; range = 60–85 years).
- *Qualitative sample:* 20 elderly participants and 10 healthcare professionals (doctors, nurses, caregivers) for in-depth interviews.
- *Inclusion criteria:* Adults aged ≥ 60 years with at least one chronic condition or prior healthcare engagement and experience with or awareness of AI-enabled or telemedicine tools.
- *Exclusion criteria:* Individuals with severe cognitive impairment preventing informed consent or participation.

Data collection

Quantitative survey

A structured questionnaire (Supplementary File S1) captured data on demographics, health conditions, awareness and use of AI tools, digital literacy, family support, and perceived empowerment. Standardized instruments included the WHOQOL-BREF [26] and the eHealth Literacy Scale (eHEALS) [17].

Qualitative interviews

Semi-structured interviews explored perceptions of AI and telemedicine, barriers to adoption, ethical concerns, and the role of family or caregivers. Interviews were conducted in participants' preferred language, each lasting approximately 20–30 min, audio-recorded with permission, and transcribed verbatim.

Data analysis

Quantitative data

Analyzed using SPSS v26. Descriptive statistics summarized participant characteristics and technology usage. Bivariate correlations assessed associations between digital literacy, family support, and empowerment. Multiple regression models identified predictors of empowerment, reporting model-fit indicators ($R^2=0.46$, adjusted $R^2=0.44$), with statistical significance set at $p<0.05$. Regression assumptions (normality, multicollinearity, homoscedasticity) were checked and satisfied.

Qualitative data

Analyzed thematically using NVivo software. An inductive coding process identified recurring themes—accessibility, usability, trust, ethics and triangulation across data sources enhanced credibility and robustness.

Ethical considerations

Written informed consent was obtained from all participants. Confidentiality and anonymity were strictly maintained throughout. The study adhered to the principles of the Declaration of Helsinki (2013 revision) and the ICMR National Ethical Guidelines (2017/2019) for social and behavioral research.

Results

Objective 1: Adoption and usage of AI-enabled telemedicine tools

Out of 200 participants, 52% reported using at least one AI-enabled healthcare tool, while 68% were aware of

such technologies. The most common applications were wearable monitoring devices (32%), teleconsultation platforms (28%), and AI-based medication-reminder apps (21%). Only 9% had experience with social or companion robots (see Fig. 1).

Adoption rates were higher among urban participants (61%) than semi-urban participants (39%), showing a significant association between place of residence and technology usage ($\chi^2=12.5$, $p<0.01$). Education level also influenced adoption—those with secondary education or higher were more likely to use AI-based tools.

Objective 2: Impact of digital literacy and family support on empowerment

Correlation analysis showed significant positive associations between digital literacy and empowerment ($r=0.48$, $p<0.05$) and between family support and empowerment ($r=0.42$, $p<0.05$). These findings indicate that individuals with stronger digital skills and family support networks feel more empowered in using health technologies (Table 1).

Regression analysis confirmed these relationships. Digital literacy ($\beta=0.41$, $p=0.002$) and family support ($\beta=0.36$, $p=0.004$) were significant predictors of empowerment, explaining nearly half the variance in empowerment scores. In contrast, age and education level were not significant predictors (Table 2).

Model Fit: $R^2=0.46$, Adjusted $R^2=0.44$. ($p<0.01$).

Regression assumptions for normality, multicollinearity, and homoscedasticity were satisfied.

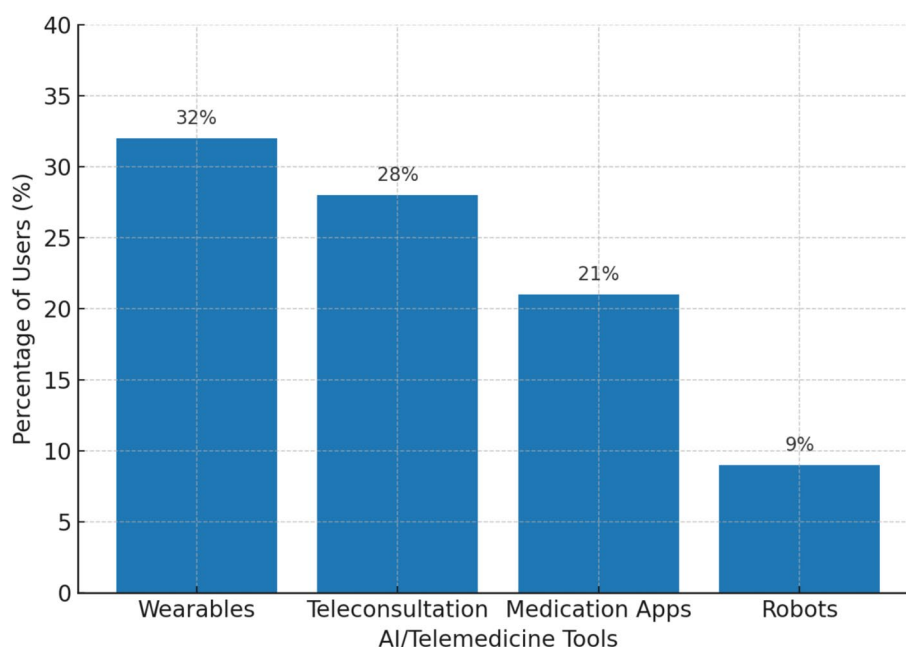


Fig. 1 Types of AI/telemedicine tools used

Table 1 Correlation matrix among key variables

Variables	Empowerment	Digital literacy	Family support	Age	Education level
1. Empowerment	1	0.48*	0.42*	−0.10	0.12
2. Digital literacy	0.48*	1	0.33*	−0.08	0.20
3. Family support	0.42*	0.33*	1	−0.05	0.09
4. Age	−0.10	−0.08	−0.05	1	−0.15
5. Education level	0.12	0.20	0.09	−0.15	1

(* $p < 0.05$)**Table 2** Regression analysis predicting perceived empowerment

Predictor	β	95% CI	p-value
Digital literacy	0.41	0.15–0.67	0.002**
Family support	0.36	0.12–0.60	0.004**
Age	−0.12	−0.31 – 0.07	0.14
Education level	0.15	−0.02 – 0.32	0.09

(** $p < 0.01$)**Table 3** Reported benefits vs. Barriers of AI-enabled elderly care

Category	Benefits	Barriers/Challenges
Elderly Adults	Convenient access; better monitoring; medication adherence	Digital literacy issues; cost; lack of trust
Healthcare Professionals	Reduced workload; efficient monitoring	Data privacy; limited training; reduced human contact

These findings underscore the pivotal role of digital inclusion and supportive family structures in promoting empowerment—factors that can offset age- or education-related disadvantages.

Objective 3: Perceived benefits and challenges(Elderly + Professionals)

Survey results indicated that the most frequently reported benefits of AI-enabled telemedicine were convenient access (61%), better chronic-disease monitoring (54%), and improved medication adherence (42%). Commonly reported challenges were low digital literacy (49%), high cost (45%), and lack of trust in AI decisions (37%). These patterns are summarized in Table 3 and visually presented in Fig. 2.

Qualitative interviews deepened these insights. Older adults valued remote consultations but often needed assistance using smartphones. One participant noted,

“I can check my blood pressure easily, but I still need my grandson to explain the readings.”

Another participant added,

“Teleconsultation saves time, but sometimes the app freezes, and I feel helpless until someone younger helps me.”

Healthcare professionals echoed these concerns, appreciating AI for efficiency but stressing human connection:

“AI tools reduce paperwork, but they can never replace the empathy older patients expect from us.”

Objective 4: Gaps and barriers to sustainable integration

Survey and interview analyses revealed systemic barriers limiting adoption among older adults (Fig. 3; Table 4). The most reported issue was equity and accessibility (55%), especially among rural or low-income respondents who cited high device costs and poor internet connectivity. The digital divide (49%) emerged as a key barrier, as many elderly participants depended on younger family members for operation and interpretation of AI tools.

Concerns about trust and human connection (46%) were frequent. Participants emphasized that technology could assist but not replace the emotional reassurance of human care. Healthcare professionals highlighted similar apprehensions, warning that over-reliance on AI may erode patient–caregiver interaction.

Ethical concerns (42%) including privacy, cybersecurity, and inadequate regulation were also prevalent. Professionals called for stronger data-protection and elder-specific regulatory frameworks.

Discussion

This mixed-methods study examined adoption patterns, predictors, and barriers related to AI-enabled telemedicine for elderly healthcare in India. Findings indicate that adoption of AI-based healthcare tools is growing but remains uneven, with greater uptake among urban and more educated older adults. Wearables and teleconsultation platforms emerged as the most commonly used technologies, while advanced AI applications and social robots had limited reach.

A key result was that digital literacy and family support were strong, independent predictors of empowerment in using AI-based healthcare tools. This finding underscores that technological inclusion depends not only on access to devices but also on the social and functional capacities that enable confident use. In collectivist cultures such as India's, where intergenerational support plays a vital caregiving role, family assistance can significantly enhance older adults' engagement with digital health tools.

At the same time, barriers including cost, low trust, privacy concerns, and reduced human interaction suggest that technology may reproduce existing inequalities if not implemented sensitively. The digital divide shaped by differences in income, education, and geography emerged

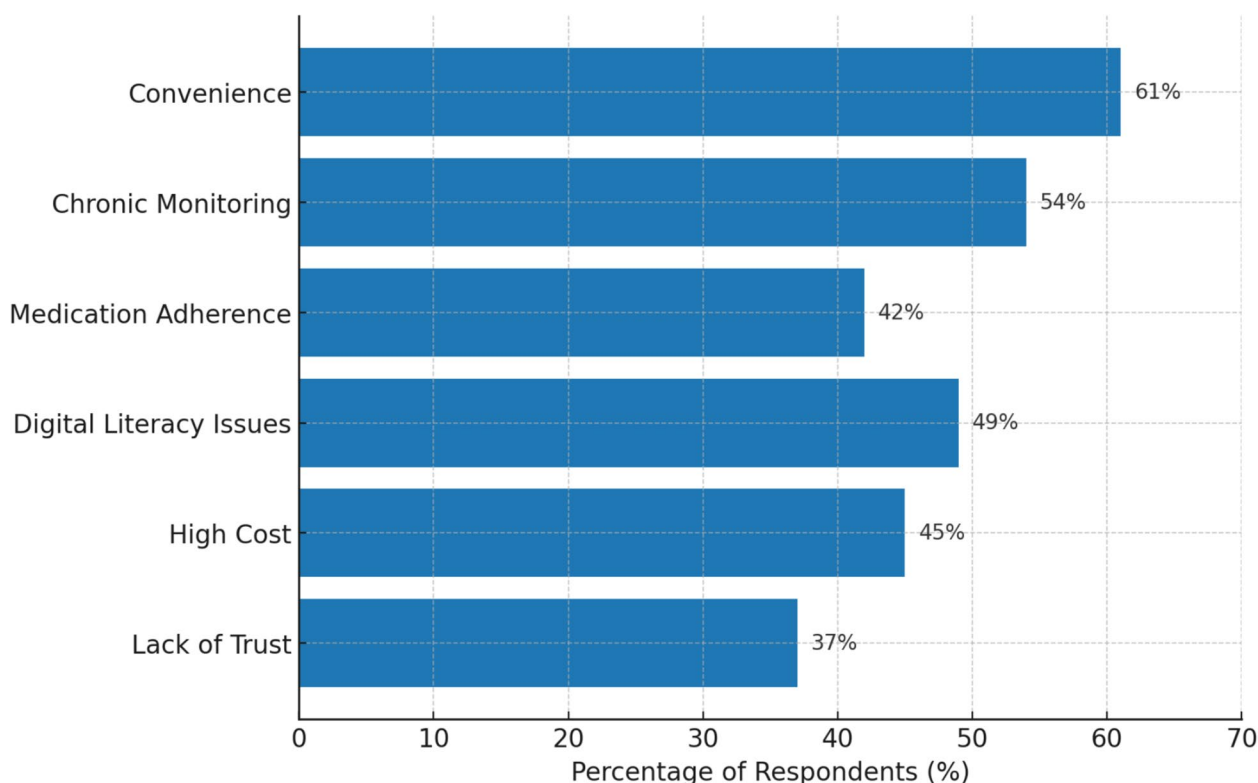


Fig. 2 Perceived benefits and challenges of AI-enabled healthcare

as a central challenge. This divide restricts meaningful participation for older adults who lack digital literacy or live in low-connectivity areas.

Geographic limitations and generalizability

While the study provides valuable insights, its geographic focus on Aligarh city limits broad generalization across India's diverse aging population. Aligarh represents an urban and semi-urban, and patterns of adoption may differ in rural or metropolitan regions with varying digital infrastructure and healthcare access. Nevertheless, the findings reflect broader trends in lower- and middle-income settings where digital exclusion remains prominent among older adults.

Linking local insights to national context

The observed barriers align with India's national efforts toward digital inclusion and telehealth expansion, such as the National Digital Health Mission (NDHM) and the Ayushman Bharat Digital Mission (ABDM). However, without age-sensitive implementation, these initiatives risk widening the gap between digitally literate and excluded elders. To bridge this divide, targeted interventions such as community-based digital literacy workshops, low-cost telemedicine kits, and caregiver-inclusive training programs should be prioritized.

Furthermore, integrating AI ethics within eldercare policy especially around privacy, consent, and algorithmic fairness can strengthen trust and encourage sustained use among older adults.

Implications for gerontology and policy

Taken together, these findings demonstrate that AI-enabled telemedicine can support empowerment, chronic-disease management, and healthcare accessibility for older adults when coupled with enabling social structures.

- *For practitioners*, training in digital communication and empathy-preserving care models can help balance technology and human touch.
- *For policymakers*, investing in digital infrastructure and senior-centered education can enhance equitable participation.
- *For researchers*, the results highlight the need for longitudinal and intervention-based studies to test causal relationships between digital literacy, family support, and empowerment.

By embedding AI and telemedicine within age-friendly policy frameworks, India can advance toward inclusive gerontechnology that honors autonomy and human connection.

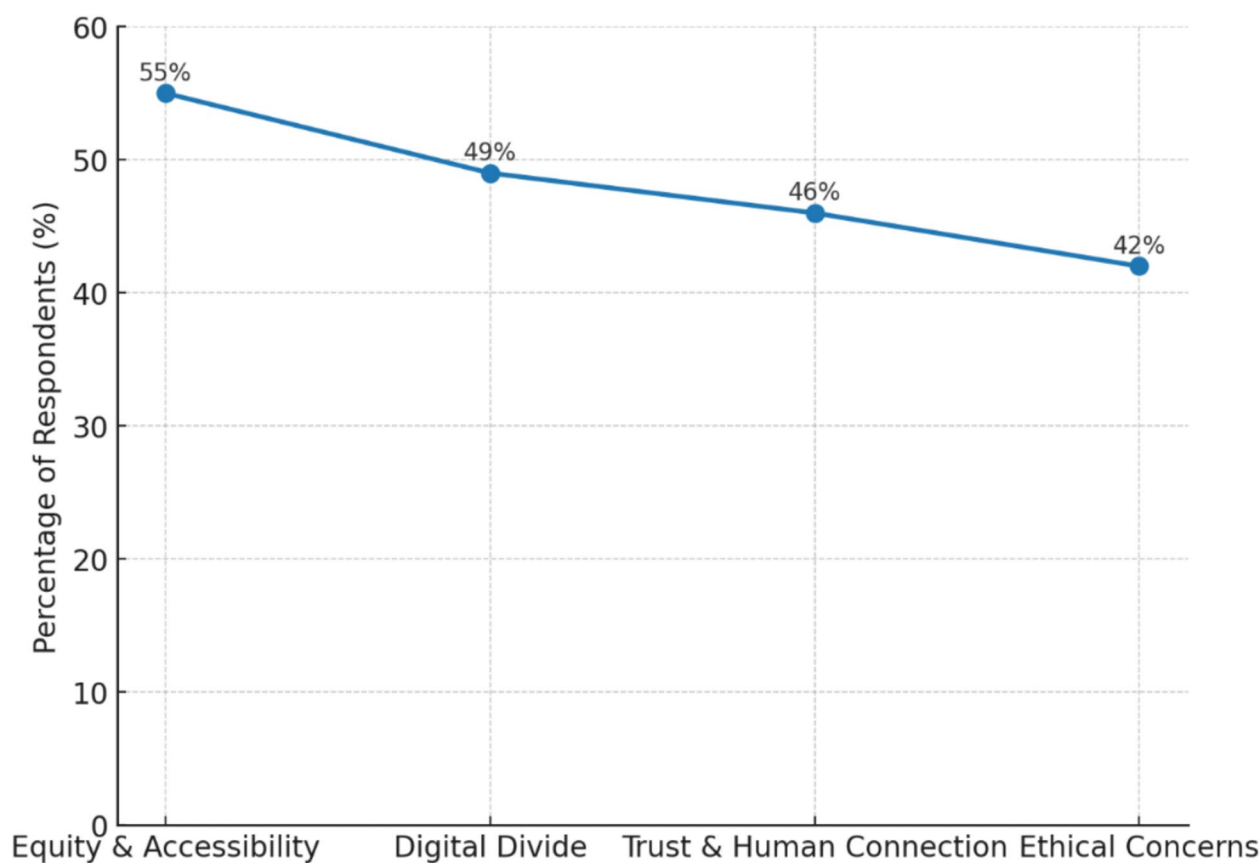


Fig. 3 Systemic barriers to AI adoption in elderly healthcare

Table 4 Systemic barriers to AI adoption in elderly healthcare

Barrier	Key issues reported	% Respon- dents
Equity & Accessibility	High cost, rural–urban disparity, internet limitations	55%
Digital Divide	Low digital literacy, dependence on younger family	49%
Trust & Human Connection	Preference for personal contact, caregiver relationships	46%
Ethical Concerns	Privacy, cybersecurity, lack of regulation	42%

Limitations

This study has several limitations. First, the sample was limited to a single region Aligarh, Uttar Pradesh which may restrict generalizability. Second, the cross-sectional design precludes causal inference about the relationships identified. Third, reliance on self-reported data may involve recall or social-desirability bias. Fourth, the structured survey did not capture long-term behavioral adoption patterns. Lastly, given the rapid pace of AI innovation, findings represent a snapshot in time.

Future research should adopt longitudinal and multi-site validation designs to examine how digital literacy and social support influence empowerment and health outcomes across different contexts and time frames.

Conclusion

This study provides evidence that AI-enabled telemedicine has the potential to improve geriatric healthcare in India by enhancing accessibility, disease monitoring, and empowerment. However, adoption remains constrained by socioeconomic inequities, digital illiteracy, and ethical challenges.

Empowerment among older adults is shaped more by digital inclusion and family support than by demographic factors such as age or education. Sustainable integration requires affordable access, digital literacy initiatives, and caregiver-inclusive support systems to ensure that technology complements rather than replaces human care.

Policy frameworks like the National Digital Health Mission should embed elder-centered strategies that promote trust, usability, and data protection. Future research must test intervention models that combine AI technologies with humanistic and culturally responsive care to achieve equitable digital health for India’s aging population.

Abbreviations

AI	Artificial Intelligence
ICT	Information and Communication Technology
HER	Electronic Health Records
IoMT	Internet of Medical Things

RCT Randomized Controlled Trial
ADL Activities of Daily Living
WHOQOL-BREF World Health Organization Quality of Life Scale–Brief
eHEALS EHealth Literacy Scale

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12877-025-06872-y>.

Supplementary Material 1.

Acknowledgements

The author gratefully acknowledges the cooperation of the elderly participants, healthcare professionals, and local organizations who assisted with data collection.

Authors' contributions

Dr. Sameen Rafi is the sole author of this study and was responsible for conceptualization, data collection, analysis, interpretation, and manuscript writing.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability

All data generated or analyzed during this study are included in this published article. Additional materials are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the ethical principles of the Declaration of Helsinki (2013 revision). According to the Indian Council of Medical Research (ICMR) National Ethical Guidelines for Biomedical and Health Research Involving Human Participants (2017; updated 2019), formal approval from an institutional ethics committee was not required for this type of non-clinical, non-invasive social research. All participants were fully informed about the objectives of the study, confidentiality and anonymity were assured, and written informed consent to participate was obtained from all participants prior to data collection.

Consent for publication

Not applicable. This manuscript does not include any individual personal data in text, images, or other formats.

Competing interests

The authors declare no competing interests.

Received: 5 September 2025 / Accepted: 28 November 2025

Published online: 16 December 2025

References

- Amini S, Zhou Y, Yao X. Speech and language biomarkers for early detection of cognitive impairment: a systematic review. *Front Aging Neurosci.* 2024;16:137492. <https://doi.org/10.3389/fnagi.2024.137492>.
- Bonanno M, Parker J, Lang C, De Rossi G. Use of wearable sensors to assess fall risk in neurological populations: a systematic review. *JMIR Mhealth Uhealth.* 2025;13(1):e67265. <https://doi.org/10.2196/e67265>.
- Birkholz L, Martin M, Barnum B, Breslin L, Kalevor S. Navigating artificial intelligence in nursing: an ethical exploration of benefits, risks, and educational shifts. *Nurs Outlook.* 2025;73(5):102520.
- Chen M, Decary M. Artificial intelligence in healthcare: an essential guide for health leaders. *Health Manage Forum.* 2020;33(1):10–8. <https://doi.org/10.1177/0840470419873123>.
- Chen Y, Wu F, Zhang W, Xing W, Zhu Z, Huang Q, et al. Perspectives on AI-driven nursing science among nursing professionals from China: a qualitative study. *Nurs Rep.* 2025;15(6):218.
- Chu CH, Donato-Woodger S, Khan SS, Nyrop R, Leslie K, Lyn A, ... Grenier A. Age-related bias and artificial intelligence: a scoping review. *Human Soc Sci Commun.* 2023;10(1):1–17.
- Follmann A, Schollemann F, Arnolds A, Weismann P, Laurentius T, Rossaint R, et al. Reducing loneliness in stationary geriatric care with robots and virtual encounters—a contribution to the COVID-19 pandemic. *Int J Environ Res Public Health.* 2021;18(9):4846. <https://doi.org/10.3390/ijerph18094846>.
- Guo Y, Hao Z, Zhao S, Gong J, Yang F. Artificial intelligence in health care: bibliometric analysis. *J Med Internet Res.* 2020;22(7):e18228. <https://doi.org/10.2196/18228>.
- Han X, Zhou X, Tan B, Jiao L, Zhang R. AI-based next-generation sensors for enhanced rehabilitation monitoring and analysis. *Measurement.* 2023;223:113758. <https://doi.org/10.1016/j.measurement.2023.113758>.
- Irfan B, Kuoppamäki S, Skantze G. Recommendations for designing conversational companion robots with older adults through foundation models. *Front Robot AI.* 2024;11:1363713.
- Lee SH, Kim J, Lim B, Lee HJ, Kim YH. Exercise with a wearable hip-assist robot improved physical function and walking efficiency in older adults. *Sci Rep.* 2023;13(1):7269.
- Li X, Zhang Q, Wu T. User acceptability of wearable fall-detection devices among older adults: a qualitative synthesis. *BMC Geriatr.* 2024;24(1):482. <https://doi.org/10.1186/s12877-024-04268-9>.
- Mao Q, Zhang J, Yu L, Zhao Y, Luximon Y, Wang H. Effectiveness of sensor-based interventions in improving gait and balance performance in older adults: systematic review and meta-analysis of randomized controlled trials. *J Neuroeng Rehabil.* 2024;21(1):85.
- Marziali E, Chou C, Sharma R. Conversational AI and social robots for older adults: a meta-analytic review of impacts on loneliness and depression. *JMIR Hum Factors.* 2024;11(2):e56789. <https://doi.org/10.2196/56789>.
- Muhammad FK, Ghazal TM, Said RA, Fatima A, Abbas S, Khan MA, et al. An IoT-enabled smart healthcare model to monitor elderly people using machine learning technique. *Comput Intell Neurosci.* 2021;2021:2487759. <https://doi.org/10.1155/2021/2487759>.
- Nair M, Svedberg P, Larsson I, Nygren JM. A comprehensive overview of barriers and strategies for AI implementation in healthcare: mixed-method design. *PLoS One.* 2024;19(8):e0305949. <https://doi.org/10.1371/journal.pone.0305949>.
- Norman CD, Skinner HA. Ehealth literacy: essential skills for consumer health in a networked world. *J Med Internet Res.* 2006;8(2):e9. <https://doi.org/10.2196/jmir.8.2.e9>.
- Pilotto A, Boi R, Petermans J. Technology in geriatrics. *Age Ageing.* 2018;47(6):771–4. <https://doi.org/10.1093/ageing/afy123>.
- Rafi S, Saif S. Age-friendly communities: creating a conducive environment for the elderly in India. *J Geriatr Med.* 2020;2(2).
- Rafi S, Saif S. (Physical and mental health issues of elderly women in India: a review. In *Handbook of Research on Policies, Protocols, and Practices for Social Work.* 2021.
- Raza H, Abbas N, Amir S, Arshad K, Siddiqui MRU, Khan SI. An IoT-enabled smart healthcare model to monitor elderly people using explainable artificial intelligence (EAI). *J NCBAE.* 2022;1(2):16–22.
- Salive ME. Multimorbidity in older adults. *Epidemiol Rev.* 2013;35(1):75–83. <https://doi.org/10.1093/epirev/mxs009>.
- Shin JH, Byeon N, Yu H, Yun G, Kim H, Lim S, et al. Effect of 4-weeks exercise program using wearable hip-assist robot (EX1) in older adults: one group pre-and post-test. *BMC Geriatr.* 2023;23(1):724.
- Talukder MSH, Nur AH, Zaman S, Noor MR, Khan MAU, Amir F. Optimizing diabetes prediction accuracy: A comprehensive approach with advanced preprocessing and diverse machine learning classifiers. In *2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE).* IEEE. 2024. pp. 1–6.
- Tan CK, Lou VW, Cheng CYM, He PC, Khoo VEJ. Improving the social well-being of single older adults using the LOVOT social robot: qualitative phenomenological study. *JMIR Hum Factors.* 2024;11(1):e56669.
- The WHOQOL Group. Development of the World Health Organization WHOQOL-BREF quality of life assessment. *Psychol Med.* 1998;28(3):551–8. <https://doi.org/10.1017/S0033291798006667>.
- Udupa P, Yellampalli SS. Smart home for elder care using wireless sensor. *Circuit World.* 2018;44(2):69–77. <https://doi.org/10.1108/CW-11-2016-0054>.

28. Wei Q, Pan S, Liu X, Hong M, Nong C, Zhang W. The integration of AI in nursing: addressing current applications, challenges, and future directions. *Front Med Lausanne*. 2025;12:1545420.
29. World Health Organization. Decade of healthy ageing: Baseline report. World Health Organization. 2021. <https://www.who.int/publications/i/item/9789240017900>.
30. Yixiao C, Hui S, Quhong S, Xiaoxi Z, Jirong Y. A review of utility of wearable sensor technologies for older person frailty assessment. *Exp Gerontol*. 2025;200:112668. <https://doi.org/10.1016/j.exger.2024.112668>.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.